

Using Richardson Extrapolation with Trinomial Trees for Pricing American Options

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Abstract

This report explores the application of Richardson extrapolation techniques to trinomial tree models for pricing American-style options. American options present unique challenges due to the possibility of early exercise, which precludes the use of closed-form solutions like the Black-Scholes formula. Trinomial trees provide a discrete-time approximation that naturally accommodates early exercise decisions. However, achieving high accuracy requires a large number of time steps, leading to computational inefficiency. Richardson extrapolation accelerates convergence by eliminating leading error terms in the asymptotic expansion of the pricing error. We discuss the theoretical foundations, implementation steps, and advantages of this approach as a benchmark for American option pricing.

1 Introduction

American-style options allow exercise at any time up to maturity, introducing the early exercise premium that complicates pricing compared to European options. The Black-Scholes-Merton framework (Black and Scholes, 1973) provides a closed-form solution for European options but cannot directly handle American options due to the optimal exercise boundary.

Lattice models, such as binomial (Cox et al., 1979) and trinomial trees (Kamrad and Ritchken, 1991), discretize the underlying asset's price dynamics and solve backwards in time, comparing continuation values with exercise payoffs at each node to account for early exercise. The Kamrad-Ritchken trinomial tree, in particular, offers improved convergence over binomial trees by allowing three possible moves (up, middle, down) per step, reducing discretization error.

However, the accuracy of tree models depends on the number of time steps N , with pricing error typically converging as $O(1/N^k)$ for some order k . For standard trinomial trees, the convergence rate is often $O(1/N^2)$ for smooth payoffs, but can be $O(1/N)$ for options with discontinuities introduced by early exercise (Moon and Kim, 2023). To achieve high precision, large N is needed, increasing computational cost.

Richardson extrapolation exploits the asymptotic error expansion to combine prices from coarse and fine grids, eliminating lower-order error terms and accelerating convergence. This technique is particularly valuable for American options, serving as a benchmark where analytical solutions are unavailable. As noted in ?, repeated Richardson extrapolation can estimate bounds on the true option value, enhancing reliability.

2 Richardson Extrapolation Technique

Richardson extrapolation is a numerical method to improve approximation accuracy by assuming an asymptotic expansion of the error. For a quantity P (e.g., option price) approximated by $P(N)$ using N steps, we assume:

$$P(N) = P + \sum_{k=1}^{\infty} \frac{A_k}{N^k},$$

where P is the true value in the continuous limit ($N \rightarrow \infty$), and A_k are unknown coefficients.

For first-order convergence ($k = 1$ dominant), a two-point extrapolation using $P(N)$ and $P(2N)$ eliminates the $O(1/N)$ term:

$$P \approx 2P(2N) - P(N) + O\left(\frac{1}{N^2}\right).$$

Derivation: Substitute the expansion:

$$P(2N) = P + \frac{A_1}{2N} + \frac{A_2}{4N^2} + O\left(\frac{1}{N^3}\right),$$

$$2P(2N) - P(N) = 2\left(P + \frac{A_1}{2N} + \frac{A_2}{4N^2}\right) - \left(P + \frac{A_1}{N} + \frac{A_2}{N^2}\right) + O\left(\frac{1}{N^3}\right) = P - \frac{A_2}{2N^2} + O\left(\frac{1}{N^3}\right).$$

For second-order ($O(1/N^2)$), the formula is:

$$P \approx \frac{4P(2N) - P(N)}{3} + O\left(\frac{1}{N^3}\right).$$

Repeated Richardson extrapolation extends this to higher orders by iteratively applying the method to multiple points (e.g., $N, 2N, 4N, \dots$). As described in ?, this involves a tableau where each column eliminates an additional error term.

A key advantage is Schmidt's inequality (Schmidt, 1968), which provides error bounds for sufficiently large N :

$$|A_{i,m+1} - P| \leq |A_{i,m+1} - A_{i,m}|,$$

allowing prediction of the true value interval and determination of required points for desired accuracy.

In the context of American options, Geske and Johnson (1984) originally applied extrapolation to Bermudan approximations (finite exercise points), but extensions to tree steps are common (Broadie and Detemple, 1996).

3 Trinomial Tree Model for American Options

The trinomial tree models the underlying asset price S under geometric Brownian motion:

$$dS = (r - q)S dt + \sigma S dW,$$

where r is the risk-free rate, q the dividend yield, σ volatility.

In the Kamrad-Ritchken model, from node S , the prices move to Su , Sm , Sd with probabilities p_u , p_m , p_d , chosen to match moments:

$$u = e^{\lambda\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad m = 1,$$

$$p_u = \frac{1}{2\lambda^2} + \frac{\sqrt{\Delta t}}{2\lambda} \left(r - q - \frac{\sigma^2}{2}\right), \quad p_d = \frac{1}{2\lambda^2} - \frac{\sqrt{\Delta t}}{2\lambda} \left(r - q - \frac{\sigma^2}{2}\right), \quad p_m = 1 - \frac{1}{\lambda^2},$$

with stretch parameter $\lambda > 1$ (often $\sqrt{3/2}$ for stability).

Pricing proceeds backwards: at maturity $T = N\Delta t$, the payoff is $\max(K - S, 0)$ for a put. At earlier nodes, the continuation value is the discounted expected value, compared to exercise value $\max(K - S, 0)$, taking the maximum for American.

Convergence studies show trinomial trees often achieve $O(1/N^2)$ for European options, but early exercise can reduce to $O(1/N)$ without smoothing (Moon and Kim, 2023). Smoothing techniques (e.g., averaging near the boundary) can restore higher order.

4 Applying Richardson Extrapolation to Trinomial Trees

For trinomial trees, Richardson extrapolation is applied to prices computed with different N . Assuming error $O(1/N)$ (conservative for American options):

1. Compute $P(N)$ with small N (fast but rough approximation).
2. Compute $P(2N)$ with doubled steps (more accurate but slower).
3. Extrapolate: $P_{\text{ext}} = 2P(2N) - P(N)$.

This yields an $O(1/N^2)$ approximation faster than direct $P(4N)$.

For higher order, use repeated extrapolation with $N, 2N, 4N$:

Assuming $P(N) = P + A/N + B/N^2 + O(1/N^3)$,

First extrapolate $P_1 = 2P(2N) - P(N) = P - B/(2N^2) + O(1/N^3)$,

Then with $P(4N)$: $P_2 = 2P(4N) - P(2N) = P - B/(8N^2) + O(1/N^3)$,

Extrapolate further: $P_{\text{ext2}} = \frac{4P_2 - P_1}{3} = P + O(1/N^3)$.

Studies confirm this accelerates trinomial convergence for American puts (Figlewski, 2009; Moon and Kim, 2023).

Using Schmidt's inequality, bounds can be estimated, e.g., $|P - P_{\text{ext}}| \leq |P_{\text{ext}} - P(2N)|$.

5 Why Richardson Extrapolation is a Good Benchmark to Replace Black-Scholes

Black-Scholes is unsuitable for American options due to early exercise. Trinomial trees with Richardson provide a rigorous numerical benchmark:

- **Handles Early Exercise:** Trees inherently optimize exercise decisions.
- **Accelerated Convergence:** Reduces computation for high accuracy, e.g., equivalent to very large N but faster.
- **Error Bounds:** Repeated extrapolation estimates true value intervals, as in ? for Bermudan but adaptable to trees.
- **Superior to Raw Trees:** Addresses non-uniform convergence issues in deep-in-the-money or high-dividend cases (Omberg, 1987).

Compared to other methods (e.g., finite differences, Monte Carlo), extrapolated trinomial trees balance accuracy and speed, serving as a benchmark in literature (Broadie and Detemple, 1996).

Simulation results in ? show repeated extrapolation reduces RMS errors significantly, applicable analogously to trees.

6 Conclusion

Richardson extrapolation enhances trinomial tree models for American option pricing by accelerating convergence and providing error estimates. This approach serves as an effective benchmark where Black-Scholes fails, rigorously handling early exercise while maintaining computational efficiency. Future work could explore adaptive λ in Kamrad-Ritchken trees with higher-order extrapolations.

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